Casual Document for Quantitative Mixing Time Results for Single Site Gibbs in Stochastic Block Model

Wen, Zehai; Cesare, Michael

2026-11-12 00:29:56-05:00

Contents

1 The Stochastic Block Model and the Single Site Gibbs Method **2**

Chapter 1

The Stochastic Block Model and the Single Site Gibbs Method

The procedure of analyzing graphical data using Bayesian stochastic block model with single site Gibbs procedure seems to have been first mentioned in Kemp et al. (2004) and slightly later in Xu et al. (2006). In this section, we describe the simplest setup of both the model and the single site Gibbs method by following a tutorial article Schmidt and Mørup (2013). We will fill in the missing steps and justify that the proposed method is indeed the standard single site systamatic scan Gibbs method.

1.1 The Bayesian Stochastic Block Model

For the moment, let us consider the simplest case where the graphical data received by the statistician is just one simple graph of $N \geqslant 3$ vertices labeled as $1, 2, \ldots, N$, represented by its adjacency matrix $X \in \{0,1\}^{N\times N}$ where $X_{i,j} = 1$ if and only if there is an edge between vertex i and vertex j. In particular, we have $X_{i,i} = 0$ for all i and $X_{i,j} = X_{j,i}$ for all i, j. If we want to, we can consider later the case which our data contains multiple graphs of the same size generated independently identically distributed from a distribution in the same model, but we shall not do so in this section.

To describe the model, some notations have to be introduced. Let us write Θ for the collection of all partitions of the set $\{1, 2, \ldots, N\}$. That is, each $z \in \Theta$ is a collection of nonempty disjoint subsets of $\{1, \ldots, N\}$ whose union equals to $\{1, \ldots, N\}$. Very frequently, we shall present an element $z \in \Theta$ by writing it as a vector $z = (z_1, z_2, \ldots, z_N) \in \{1, \ldots, N\}^N$ where z_i denotes the block to which vertex i belongs. For example, if $N = 5$, both $z = (1, 2, 1, 3, 3)$ and $z = (5, 1, 5, 2, 2)$ correspond to the partition $z = \{\{1, 3\}, \{2\}, \{4, 5\}\}$. z is said to be a *restricted growth string* if the blocks are ordered as they first appear: $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, we have

$$z_i \in \left\{1, 2, \ldots, 1 + \max_{j<i} z_j\right\}.$$

In the previous example, $(1, 2, 1, 3, 3)$ is a restricted growth string and the other is not.

We assume that X is generated from a distribution in the (standard) stochastic block model (SBM) with unknown number of blocks. The model contains two parameters: the partition z ranged over Θ and the connectivity probabilities $\phi_{k,l} \in [0, 1]$, where $k \in z$ and $l \in z$ are the communities. The model assumes that, given z and ϕ, independently over all other edges, an edge is joined between a vertex in block k and a vertex in block l with probability ϕ_{kl}. In symbols:

$$p(X \mid z, \phi) = \prod_{1\leqslant i<j\leqslant N} \phi_{z_i,z_j}^{x_{ij}}(1 - \phi_{z_i,z_j})^{1-x_{ij}}, \tag{1.1}$$

where we have written $p\left(X \mid z, \phi\right)$ in place of $p\left((X_{i,j})_{1 \leqslant i < j \leqslant N} \mid z, \phi\right)$. Another look at (1.1) supply a second formula:

$$p\left(X \mid z, \phi\right) = \prod_{k,l \in z} \phi_{k,l}^{m_{k,l}} (1 - \phi_{k,l})^{\bar{m}_{k,l}}, \tag{1.2}$$

where $m_{k,l}(\bar{m}_{k,l})$ is the number of edges(non-edges) between block k and block l, and, here and below, $\prod_{k,l \in z}$ runs over unordered pairs $\{k, l\}$ of blocks, including $k = l$.

In order to do Bayesian statistics, priors need to be posed for z and for ϕ. The standard practice for this model, as discussed in Schmidt and Mørup (2013), is to first implement the conjugate prior for ϕ, integrate ϕ out to get a formula for $p\left(X \mid z\right)$, treat the model as if z is the only parameter, give z a prior distribution, and then carry out Markov chain Monte Carlo(MCMC) on the posterior distribution $p(z \mid X) \propto p(X \mid z)p(z)$. This is the procedure we will follow.

With hyperparameters $a > 0$ and $b > 0$ fixed by the statistician, we require all $\phi_{k,l}$'s to have a prior distribution of Beta(a, b), independently with respect to each other and over all other variables. Consequently:

$$\begin{aligned} p(X \mid z) &= \int_{[0,1]^{\frac{|z|(|z|+1)}{2}}} p(X \mid z, \phi) p(\phi) \mathrm{d}\phi = \int_{[0,1]^{\frac{|z|(|z|+1)}{2}}} \prod_{k,l \in z} \frac{1}{B(a,b)} \phi_{k,l}^{m_{k,l}+a-1} (1 - \phi_{k,l})^{\bar{m}_{k,l}+b-1} \mathrm{d}\phi \\ &= \prod_{k,l \in z} \frac{1}{B(a,b)} \int_0^1 \phi_{k,l}^{m_{k,l}+a-1} (1 - \phi_{k,l})^{\bar{m}_{k,l}+b-1} \mathrm{d}\phi_{k,l} \\ &= \prod_{k,l \in z} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a,b)}, \end{aligned} \tag{1.3}$$

where $B(\cdot, \cdot)$ is the beta function. From now on, we treat z as the only parameter of the model, defining a density of X via (1.3). As justified in Schmidt and Mørup (2013), a commonly used prior distribution for z is known as the *Chinese restaurant process (CRP)*. With a hyperparameter $A > 0$ fixed by the statistician, CRP(A) is a probability measure on Θ defined as follows: with restricted growth notation, $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, given $z_1, z_2, \ldots, z_{i-1}$ who have already used a total of $K^{(i-1)}$ blocks, z_i will join an existing block $k \in \{1, 2, \ldots, K^{(i-1)}\}$ with probability

$$\frac{n_k^{(i-1)}}{A + i - 1},$$

where $n_k^{(i-1)}$ is the number of vertices among $1, 2, \ldots, i-1$ who have been assigned to block k, and will create a new block $K^{(i-1)} + 1$ with probability

$$\frac{A}{A + i - 1}.$$

CRP(A) does have a clean joint density (with respect to the counting measure on Θ)

$$p(z) = \frac{A^{|z|}\Gamma(A)}{\Gamma(A + N)} \prod_{k \in z} \Gamma(n_k), \tag{1.4}$$

where n_k is the number of vertices assigned to block k according to z, and $\Gamma(\cdot)$ is the gamma function.

By Bayes's theorem, according to (1.3) and (1.4), the posterior distribution of z given X is

$$p(z \mid X) \propto p(X \mid z)p(z) \propto \left[\prod_{k,l \in z} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a,b)}\right] \left[A^{|z|} \prod_{k \in z} \Gamma(n_k)\right]. \tag{1.5}$$

We remark that the $B(a, b)$ term in the denominator should not go to the normalizing constant: it gets multiplied by $|z|(|z| + 1)/2$ times.

1.2 The Single Site Systematic Scan Gibbs Suggested in Schmidt and Mørup (2013)

We now present the single site systematic scan Gibbs method with the goal being sampling from (1.5).

1.3 Justifications

We now justify that the method presented in Section 1.2 indeed corresponds to the single site systematic scan Gibbs sampler in a slightly different space.

Classically speaking, the single site systematic scan Gibbs sampler refers to a solution of the following problem: suppose that we wish to sample from a probability measure $\pi(\theta)$ on a σ-finite space (Θ, F_Θ, μ) defined by a desnity $\mathrm{d}\pi(\theta) = p(\theta)\mathrm{d}\mu(\theta)$. If it happens that Θ is a product space (i.e. $\Theta = \Theta_1 \times \cdots \Theta_N$ and $\mu = \mu_1 \times \cdots \times \mu_N$) and the full conditional distributions $p(\theta_i \mid \theta_{(-i)})$, where $\theta_{(-i)} = (\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_N)$, can be sampled, then the *single site systematic scan Gibbs sampler* samples π by iteratively sampling at each coordinate and standard theory covers its qualitative convergence results Robert and Casella (2004).

The problem here is that, when Θ is the collection of all partitions of $\{1, \ldots, N\}$, it is entirely unclear how Θ can be written as a product space $\prod_{i=1}^N \Theta_i$, where Θ_i corresponds to the vertex i. If we simply represent each $z \in \Theta$ using the restricted growth notation, then the vector $z_{(-i)}$ obtained by simply cropping the ith entry is no longer in the restricted growth notation and further analysis based on this notation is invalid.

(Here is a clean and slightly annoying solution I was able to find.)

Instead of working on Θ, we work on an augmented version of it. Let us define:

$$\tilde{\Theta} = \prod_{i=1}^N \tilde{\Theta}_i = \{1, \ldots, N\}^N,$$

$F_{\tilde{\Theta}} = 2^{\tilde{\Theta}}$, and μ_i be the counting measure on $\tilde{\Theta}_i = \{1, \ldots, N\}$. For each $c = (c_1, \ldots, c_N) \in \tilde{\Theta}$ and $i \in \{1, \ldots, N\}$, let $c_{(-i)} = (c_1, \ldots, c_{i-1}, c_{i+1}, \ldots, c_N)$ really be the vector cropping out the ith coordinate. For each $l \in \{1, \ldots, N\}$, let us write $(l, c_{(-i)})$ for the vector $(c_1, \ldots, c_{i-1}, l, c_{i+1}, \ldots, c_N)$, filling l to the ith place. A map $\Pi : \tilde{\Theta} \to \Theta$ can be defined by setting $\Pi(c)$ to be the partition which c represents, in the way discussed as in Section 1.1. If $p(z)$ is any density on Θ (with respect to the counting measure), then

$$\tilde{p}(c) = \frac{(N - |\Pi(c)|)!}{N!} p(\Pi(c))$$

is a density with respect to μ on $(\tilde{\Theta}, F_{\tilde{\Theta}}, \mu)$ with $C \sim \tilde{p}$ implying $\Pi(C) \sim p$.

Let us reformulate the model with parameter space being the augmented space. In this way, the model is no longer identifiable and (1.3) now reads

$$p(X \mid c) = \prod_{k,l \in \Pi(c)} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)}. \tag{1.6}$$

The prior (1.4) now becomes

$$\tilde{p}(c) = \frac{(N - |\Pi(c)|)!}{N!} \frac{A^{|\Pi(c)|}\Gamma(A)}{\Gamma(A + N)} \prod_{k \in \Pi(c)} \Gamma(n_k). \tag{1.7}$$

The posterior (1.5) now becomes

$$\tilde{p}(c \mid X) \propto \frac{(N - |\Pi(c)|)!}{N!} \left[\prod_{k,l \in \Pi(c)} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)} \right] \left[A^{|\Pi(c)|} \prod_{k \in \Pi(c)} \Gamma(n_k) \right]. \tag{1.8}$$

We now proceed to compute the full conditional probabilities. Given the clear product space structure, we separate $c = (c_i, c_{(-i)})$ without worry. We have, for each $l \in \{1, \dots, N\}$:

$$\tilde{p}(c_i = l \mid c_{(-i)}, X) = \frac{\tilde{p}(X \mid c_i = l, c_{(-i)})\tilde{p}(c_i = l \mid c_{(-i)})\tilde{p}(c_{(-i)})}{\tilde{p}(c_{(-i)}, X)} \propto \tilde{p}(X \mid c_i = l, c_{(-i)})\tilde{p}(c_i = l \mid c_{(-i)}).$$

Therefore, the only thing left is to compute these two terms.

To compute $\tilde{p}(c_i = l \mid c_{(-i)})$, we use the joint distribution $\tilde{p}(l, c_{(-i)})$ given in (1.7), which requires some notations. Let us write $\Pi(c_{(-i)})$ for the partition of the set $\{1, \dots, i-1, i+1, \dots, N\}$ obtained by starting with $\Pi(c)$ and deleting vertex i (if the action creates an empty block, delete the empty block as well) and, for each $k \in \Pi(c_{(-i)})$, $n_k^{(-i)}$ for the number of vertices in block k. It is clear that $\sum_{k \in \Pi(c_{(-i)})} n_k^{(-i)} = N - 1$ because the vertex i is not in any $k \in \Pi(c_{(-i)})$. We now discuss by cases. If the choice $c_i = l$ create a new block from $\Pi(c_{(-i)})$ (that is, $\left|\Pi(l, c_{(-i)})\right| = \left|\Pi(c_{(-i)})\right| + 1$), then

$$\tilde{p}(l, c_{(-i)}) = \frac{(N - \left|\Pi(c_{(-i)})\right| - 1)!}{N!} \frac{A^{\left|\Pi(c_{(-i)})\right|+1}\Gamma(A)}{\Gamma(A+N)} \times \Gamma(1) \times \prod_{k \in \Pi(c_{(-i)})} \Gamma(n_k^{(-i)}) = \frac{A}{N - \left|\Pi(c_{(-i)})\right|} q(c_{(-i)}),$$

where

$$q(c_{(-i)}) = \frac{(N - \left|\Pi(c_{(-i)})\right|)!}{N!} \frac{A^{\left|\Pi(c_{(-i)})\right|}\Gamma(A)}{\Gamma(A+N)} \prod_{k \in \Pi(c_{(-i)})} \Gamma(n_k^{(-i)}).$$

Secondly, suppose that the choice $c_i = l$ enters an existing block from $\Pi(c_{(-i)})$, say inside a block $r \in \Pi(c_{(-i)})$. In this case, we have $\left|\Pi(l, c_{(-i)})\right| = \left|\Pi(c_{(-i)})\right|$,

$$\tilde{p}(l, c_{(-i)}) = \frac{(N - \left|\Pi(c_{(-i)})\right|)!}{N!} \frac{A^{\left|\Pi(c_{(-i)})\right|}\Gamma(A)}{\Gamma(A+N)} \times \Gamma(n_r^{(-i)} + 1) \times \prod_{k \in \Pi(c_{(-i)}), k \neq r} \Gamma(n_k^{(-i)}) = n_r^{(-i)} q(c_{(-i)}).$$

We can thus combine the two cases and observe that, in the formula of $\tilde{p}(l, c_{(-i)})$, the block which l belongs will determine what is multiplied to $q(c_{(-i)})$ and this is the only thing that depends on l. We thus conclude that

$$\tilde{p}(c_i = l \mid c_{(-i)}) = \frac{1}{A + N - 1} \begin{cases} n_r^{(-i)} & l \text{ enters block } r \in \Pi(c_{(-i)}), \\ \frac{A}{N - \left|\Pi(c_{(-i)})\right|} & l \text{ creates a new block.} \end{cases} \tag{1.9}$$

It can be verified directly that this is a proper density:

$$\sum_{l=1}^{N} \tilde{p}(c_i = l \mid c_{(-i)}) = \frac{1}{A+N-1} \left(\left(N - \left|\Pi(c_{(-i)})\right|\right) \frac{A}{N - \left|\Pi(c_{(-i)})\right|} + \sum_{r \in \Pi(c_{(-i)})} n_r^{(-i)} \right) = \frac{A+N-1}{A+N-1} = 1.$$

To compute $\tilde{p}(X \mid c_i = l, c_{(-i)})$, we use (1.6) and again discuss by cases. We introduce a bit more notations. First, let $X_{(-i)}$ be the graph deleting the vertex i along with edges connecting i to other vertices (in the density formula, this simply stands for all edges and non-edges with ends not being i) and let

1.4 Next Steps

To proceed

References

Charles Kemp, Thomas L. Griffiths, and Joshua B. Tenenbaum. Discovering latent classes in relational data. AI Memo 2004-019, Massachusetts Institute of Technology, Computer Science and Artificial Intelligence Laboratory, Cambridge, MA, September 2004. URL http://hdl.handle.net/1721.1/30489. Also issued as MIT-CSAIL-TR-2004-050.

Christian P. Robert and George Casella. *Monte Carlo Statistical Methods*. Springer Texts in Statistics. Springer, New York, second edition, 2004. ISBN 0-387-21239-6.

Mikkel N. Schmidt and Morten Mørup. Nonparametric Bayesian modeling of complex networks: An introduction. *IEEE Signal Processing Magazine*, 30(3):110–128, May 2013. doi: 10.1109/MSP.2012.2235191.

Zhao Xu, Volker Tresp, Kai Yu, and Hans-Peter Kriegel. Infinite hidden relational models. In Rina Dechter and Thomas S. Richardson, editors, *Proceedings of the Twenty-Second Conference on Uncertainty in Artificial Intelligence (UAI 2006)*, pages 544–551, Arlington, VA, 2006. AUAI Press. URL https://arxiv.org/abs/1206.6864.